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Comparative Evaluation of Inertial Measurement Unit Versus Vision Based Motion Capture

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Abstract—Accurate motion capture is critical for biomechanics research, clinical gait analysis, human—computer interaction, and the entertainment industry. Two dominant paradigms exist: marker-based systems, which use inertial measurement units affixed to the body, and marker-less, vision-based systems, which infer kinematics directly from camera data using deep-learning techniques. While both have matured rapidly, a systematic, head-to-head evaluation under identical experimental conditions is still lacking. This study benchmarks the spatiotemporal accuracy, robustness, and practical usability of state-of-the-art inertial measurement units and vision-based pipelines across a representative set of motor tasks (walking, running, squatting, and upper-limb reaches) performed by 5 healthy adults. Joint angles from inertial measurement units were reconstructed via a complementary-filter fusion of accelerometer, gyroscope, and magnetometer signals. Vision-based approaches, specifically the most advanced methods, rely on cameras that capture images in red, green, and blue color channels and employ a series of convolutional layers for reconstruction. Additionally, we have also included double, depth-based Kinect version two cameras for comparison. Results indicated that both the inertial measurement unit and vision-based pipelines estimated lower-limb joint angles with generally acceptable differences, yet the vision approach showed larger deviations, especially during out-of-plane motion and brief self-occlusions. Inertial measurement unit, by contrast, remained more consistent in those scenarios but exhibited gradual drift during prolonged recordings when magnetic updates were absent. Although the camera-only system offered quicker setup and greater participant comfort, a streamlined calibration routine narrowed the preparation gap for inertial measurement units. Vision hardware was less expensive, but its higher computational demands offset that advantage. Taken together, the study highlights a trade-off: marker-less vision prioritizes plug-and-play usability, whereas inertial measurement units deliver steadier, higher-precision tracking.

Keywords: Human Computer Interaction, Kinect, motion capture, Dynamic Time Warping, Convolution Neural Networks, action recognition

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INTRODUCTION

Human-pose estimation and motion-capture (MoCap) technologies have become foundational to a wide array of disciplines, powering everything from ergonomic risk screening on factory floors [6] and objective rehabilitation in clinics [19] to cinematic character animation [15], sports-performance analytics [31], and the perception—action loops of collaborative robots. To translate real-world movements into analyzable kinematic data, engineers draw on several sensing paradigms. Wearable inertial-measurement-unit (IMU) arrays infer segment orientations from accelerometer and gyroscope signals [23] and thus work reliably in cluttered or outdoor settings, but require careful calibration and can drift over time [7]. Marker-less vision pipelines, propelled by convolutional neural networks that detect keypoints directly in RGB frames [29], promise unobtrusive "plug-and-play" deployment with commodity cameras yet remain sensitive to occlusions, lighting, and multi-person ambiguity [5]. Depth cameras such as Microsoft's Kinect occupy a middle ground: they fuse structured-light or time-of-flight depth maps with learned skeletal priors [25], yielding richer three-dimensional information than monocular RGB while avoiding body-worn hardware-but still struggle when limbs self-occlude or subjects face away from the sensor [21]. Because small angular errors can propagate into incorrect musculoskeletal-disorder (MSD) risk scores, misguide surgical-planning tools, or cause perceptible

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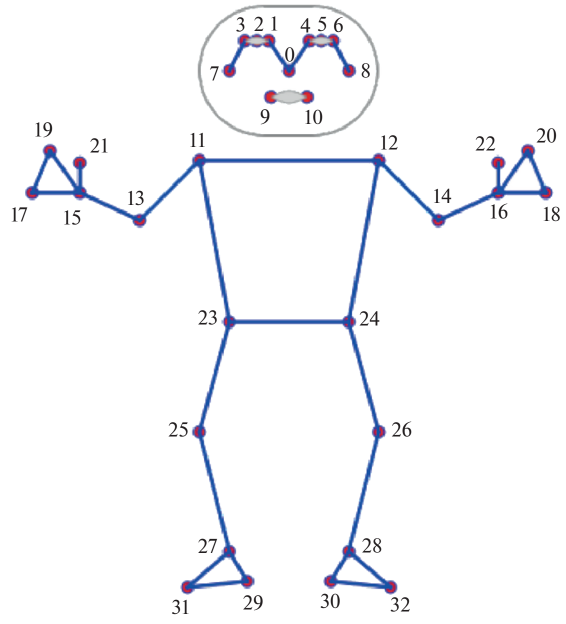


Fig. 1. OpenPose 33 keypoints topology.

artifacts in real-time applications [10], systematic accuracy research and cross-modal benchmarking are indispensable. Only by rigorously quantifying the limits and failure modes of each modality can researchers, clinicians, and industry practitioners make informed choices and ensure that MoCap-derived insights translate into safe, effective, and economically viable solutions.

VISION BASED MOTION CAPTURE APPROACHES

CNN Based Motion Capture

As category name suggests the input in this case is one image or the sequence of images from single RGB camera. What makes the task of pose estimation especially challenging is the variety of human poses, parts of human that may be covered and presence of unknown number of people at any scale in frame. Challenges during 3d reconstruction involve depth ambiguity of monocular data, multiple potential 3d poses may be mapped to the same 2d pose. Most competitive techniques suggest that usage of multiple stacked convolutional networks can achieve real-time 2d pose estimation with small error rate considering that dominant part of body is visible. State of art approaches are using sequence of 2d poses as an input and use transformer and attention mechanisms for final reconstruction. Reconstruction from sequential data shows significant improvement in accuracy.

Pose estimation in 2D space is relatively well studied problem in computer science and has various applications including hand gesture tracking, activity tracking etc.

Wide known approach for tackling this issue is to construct active zones heatmap for 17 human pose landmarks [27]. This approach involves constructing deep convolutional neural networks and implementation of high-resolution networks (HRNet) which consist of high res. networks at first stage and high to low resolution networks which eventually form the output heatmap. While heatmap based approaches and HRNet in particular has shown real-time results at average machine, they luck speed at mobile devices instead regression based methods which are slightly more accurate but give improvements in performance were used for this matter, Newell et al. [17] has used stacked hourglass architecture and has showed that it performs much faster and outputs more precise results even with smaller number of parameters. This architecture was furtherly improved by Bazarevsky et al. [2] who has added heatmap branch discard functionality which makes entire process significantly lightweight and make it possible to run on mobile devices. Inference model involved by Bazarevsky et al. [2] is a detector-tracker setup. When detector handles human landmarks detection and tracker track's its movement over frames. Detector reiterates over frames again when landmarks tracking is lost.

Per observations it is assumed that strongest signal about the position of the torso is humans face, therefore Bazarevsky et al. used face described at [3] as a proxy for person detector. Face detector used additionally predicts alignment parameters such as: the middle point between the person's hip, approximated the size of human body and the angle between the lines connecting mid shoulder and mid-hip points, see Fig. 1. Topology used is shown at Fig. 1.

Training dataset consists of 60 K images with single or multiple people at casual poses and 25 K images with single person performing fitness exercises. All images are annotated manually.

Pose predictor, which estimates the location of all 33 keypoints, uses combined heatmap, offset and regression approach (Fig. 3). Combined heatmap, offset and regression approach was adopted (Fig. 3). Heatmap and offset loss is used only at training stage and corresponding output layers are removed from the model before running the inference. Therefore, using heatmap to supervise the lightweight embedding, which is then utilized by regression encoder network. Approach used by Bazarevsky et al. [2] has been partially inspired by Stacked Hourglass method inspired by Newell et al. [17] with main difference that they also stack encoder-decoder heatmap-based network followed by a regression encoder network.

For purpose of balance achievement skip connections at all stages of the network are used. Heatmap improvement and coordinate regression accuracy was

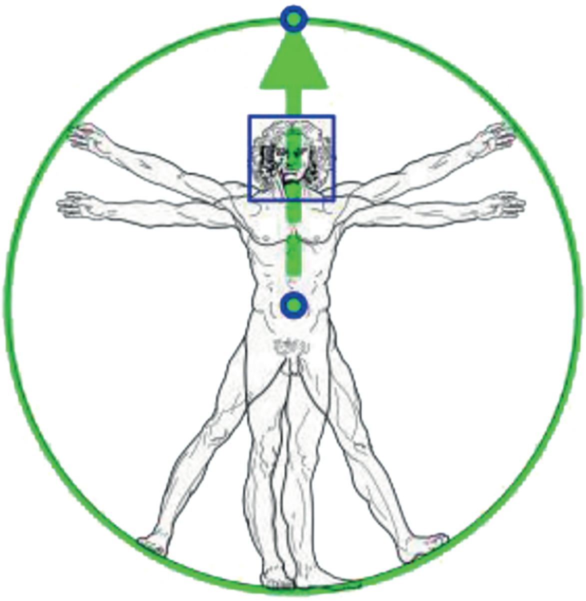


Fig. 2. Vitruvian man aligned via our detector vs. face detection bounding box.

achieved by removing gradients from regression encoder to heatmap-trained feature maps.

As part of performance improvements during training stage limited range for joint angle, scale and translation is used.

After initial detection or previous frame data, alignment operation was is done in such a way that the point between the hips is located at the center of the square image which is passed as the neural network input. Rotation is estimated as the line L between mid-hip and mid-shoulder points and rotate the image, so L is parallel to the y-axis. For cases when person has significant occlusion or part of body is not visible occlusion is being simulated during training and per point visibility classifier is used.

To evaluate our model's quality, OpenPose [5] was used as a baseline. Although models show slightly worse performance than the OpenPose model on the AR dataset, BlazePose Full outperforms OpenPose on Yoga/Fitness use cases. At the same time, BlazePose performs 25–75 times faster on a single mid-tier phone CPU.

Motion Capture Using Kinect and iPi Software

Markerless motion capture using Microsoft Kinect and software solutions like iPi Motion Capture has gained significant traction as an accessible alternative to traditional marker-based systems, vastly due to its affordability, ease of setup, and portability [25]. Figure 1 illustrates a typical Kinect sensor (Kinect v2/Azure Kinect with time-of-flight technology), which simultaneously captures RGB and depth data for real-time skeletal tracking.

The Microsoft Kinect v2 (Fig. 4a) combines an RGB camera, an infrared (IR) camera, and three IR projectors. Depth is measured via a time-of-flight (ToF) approach. The RGB sensor captures 1920 × 1080 px images, while the IR sensor captures 512 × 424 px images; both stream at 30 Hz. Through the Kinect for Windows SDK 2.0, developers can track up to six people simultaneously, with 25 skeletal joints per person. For every joint, the SDK outputs both its three-dimensional position and orientation (as a quaternion). The optical center of the IR camera lens defines the origin of the device's 3D coordinate system, which is also used for skeleton tracking (Figure 4b). Table 1 offers a detailed comparison of these specifications [1].

Various research has found it beneficial to use this technology in biomechanics, rehabilitation, and

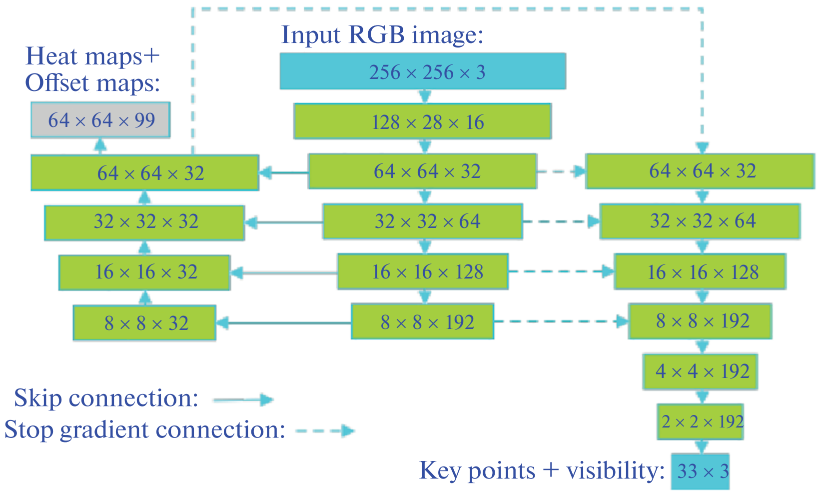
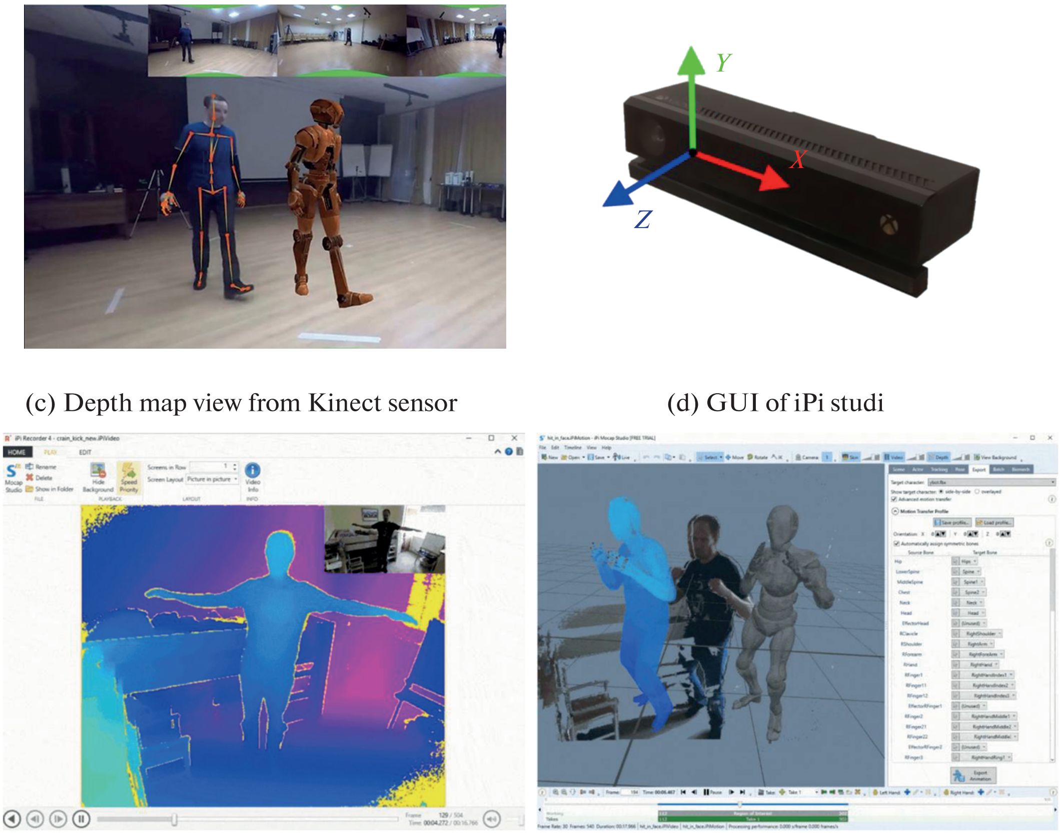


Fig. 3. Network architecture.



(a) Multiview, 3 Kinect sensors setup, with output.

(b) Kinect V2 sensor

(c) Depth map view from Kinect sensor

(d) GUI of iPi studi

Fig. 4. Kinect V2 sensor and iPi studio.

human–computer interaction, leveraging its simplified workflow: after positioning one or multiple Kinect sensors around the performer.

One notable software solution that we have employed for testing purposes is iPi software Figs. 4c, 4d. It uses silhouette-based tracking to reconstruct a 3D skeleton offline, smoothing many of the real-time processing constraints of other systems. In iPi's offline pipeline, tracking is driven by advanced silhouette-based algorithms that perform iterative error minimization: a virtual skeleton model is superimposed on the depth data from each viewpoint, and the system refines limb positions frame by frame by comparing the difference (or "error") between the model's projected silhouette and the real performer's silhouette.

Table 1. Kinect V2 hardware properties

RGB Camera Resolution	1920 × 1080 px
IR Camera Resolution	512 × 424 px
Framerate	30 fps
Field of View IR Camera	70 × 60 deg
Field of View RGB Camera	84 × 53 deg

This process typically incorporates inverse kinematics constraints to maintain realistic joint motion while reducing noise and correcting partial occlusions [30]. Figure 4 shows a multi-sensor capture setup, which is beneficial when a single sensor's view of specific limbs becomes obscured [13] Multiple Kinect sensors provide overlapping fields of view that enhance data completeness and minimize erroneous pose estimates, especially for fast or complex movements.

Nonetheless, environmental factors (e.g., reflective surfaces, sunlight) and the sensor's limited depth resolution can still introduce inconsistencies. Figure 4d depicts the iPi interface, where camera calibration (using a reference board or wand) defines each Kinect's intrinsic and extrinsic parameters before tracking.

In our double-Kinect multiview setup, we first perform intrinsic calibration of each depth sensor to obtain K_i [32] and estimate extrinsic parameters (R_i, t_i) by solving the Perspective-n-Point problem with RANSAC (Random Sample Consensus). For each pixel (u, v) in the depth map $D_i(u, v)$, we back-project into camera coordinates via

$$P_i = D_i(u,v)\,K_i^{-1}\begin{bmatrix}u\\v\\1\end{bmatrix}$$

and then transform into the shared world frame: $X_i = R_iP_i + t_i$. Given detected 2D joint locations $\{(u_{ij}, v_{ij})\}$ across the two views, we recover each 3D joint J_j by minimizing the reprojection error

$$\mathbf{J_j} = \arg\min_X \sum_i^2 \left\|\prod\left(K_i\left[R_iX + t_i\right]\right) - u_{ij}\right\|^2,$$

where $\prod\left(\left[x,y,z\right]^T\right) = (x/z, y/z)$ and this Direct Linear Transformation (DLT) system is solved as in Hartley and Zisserman (2003) [11]. Finally, to enforce kinematic consistency and suppress sensor noise, the set of all joints $\{J_j\}$ is refined via a bundle-adjustment optimization with soft anatomical priors [28] and an ICP-based alignment step. This pipeline extends the real-time pose estimation paradigm introduced by Shotton et al. [25] and incorporates depth-accuracy models from Khoshelham and Elberink [12].

As depth-sensing hardware advances and machine learning—based pose estimation evolves, the Kinect—iPi framework remains a compelling solution for robust, budget-friendly motion capture across a variety of application domains.

MARKER-BASED RECONSTRUCTION WITH IMU SENSORS

Most straight forward solutions for pose and joint coordinate tracking suggest that markers can be attached to the body joints and further track of coordinates of these markers can efficiently track motion over time [8]. Types of markers can also vary based on the activity performed or by a number of persons we try to capture. One of the available solutions for marker-based motion capture involves usage of so-called IMU (inertial measurement unit) markers. They are equipped with different types of sensors including gyroscope, accelerometer, magnetometer, transmitter etc. Each marker requires a source of power to operate and transmit information about their location to the receiver. Transmission of information is usually done with the WIFI network between computers and markers, in this case they need to share a common network. IMU based approaches don't require a specific studio for recording of the movements and this type of hardware is considered as portable therefore ease to setup, transfer, and use [14]. Things to be considered here is that IMU uses a magnetometer therefore its signal can be easily distracted by metallic objects nearby [20]. Another shortfall of IMU is lack of proper information regarding movement of subjects in space which can cause sliding issues of the subject's skeleton and incorrect capture of coordinates of the root of the skeleton.

We have used Rokoko Smartsuit Pro Fig. 5, which is wearable suit containing an array of IMU sensors often used for movie production. Suite is equipped with 19 IMU markers [14] that are able to stream data at 100 frames per second rate. Locations of sensors is shown at Fig. 6. For powering sensors, external battery needs to be attached. Each sensor is connected to wireless communication hub with cables. Hub is powered by external battery, 5000 mAh battery which allows 6 h operation time. After brief position calibration, motion capture can be started. This type of solution is portable, low-cost and easy to implement as it consists of small components. Primary limitations associated with this type of approach include limited sensors precision and influence of environment conditions.

Limited precision of the sensors affects overall motion capture accuracy causing skeleton drifting and leading to consequential measurements errors. Furthermore, IMUs rely on accelerometer and magnetometer sensors are influenced by environment conditions, such as magnetic inference. Related software for motions capture known as Rokoko Studio notifies about magnetic inference if any exists. In addition, it hosts various tools for processing of captured data. Including tools for trimming data and filters to reduce movement errors and noise leading to overall smoother data. Further positional errors, if exist, can be fixed with postprocessing with 3rd party software. We have used Blender's built in nonlinear animation panel for this aim.

Typical IMU Fig. 6, consists of combination of accelerometer, gyroscope and magne-tometer [18]. Combined data provided by these devices is then processed and used for accurate reconstruction of the IMU movement. Assuming that device is attached to the join and is matching with the actual movement of the joint of the human's body, our problem can be formulated as a reconstruction of the sequence of x, y and z coordinates relative to the root position.

Additionally for sake of data transmission IMUs are equipped with Wi-Fi SoC (ESP8622-12E) Fig. 7 which rely on I2C protocol for data transmission and communication with gyroscope, accelerometer and magnetometer [18]. The angular rate (wx; wy; wz), from the gyroscope is integrated to get the angular position while the data from the accelerometer and magnetometer is used to correct the drift caused due to the gyroscope bias and noise. This data is sent to a UDP server running in the computer software, game engine to obtain a real-time animation of the avatar. The attitude estimation algorithm runs at around 600 Hz and the animation is displayed at 20FPS.

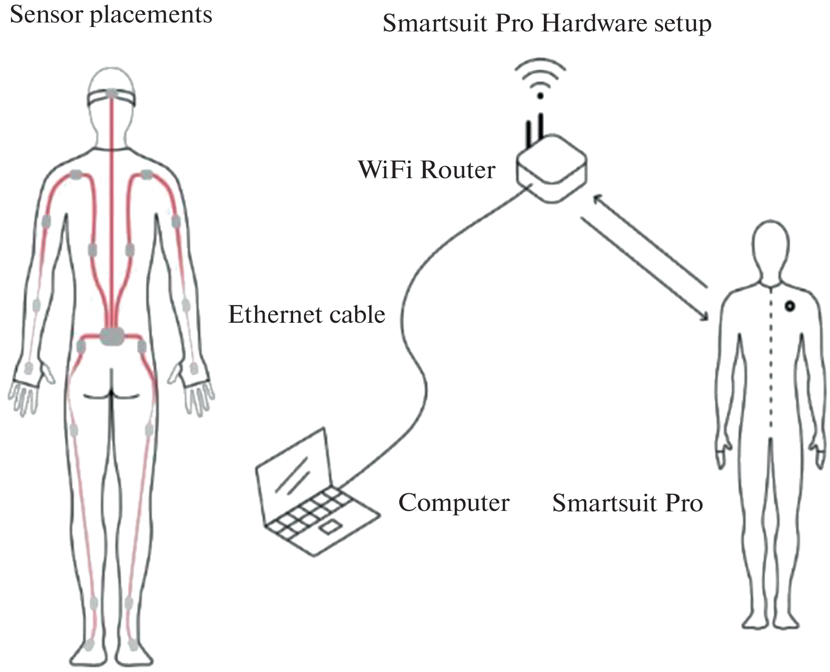


Fig. 5. Rokoko Smart Suite Pro operation scheme.

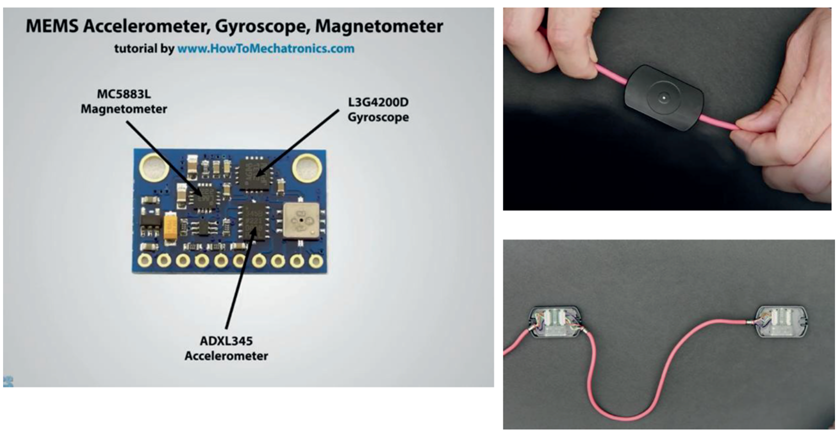


Fig. 6. IMU sensor.

EXPERIMENTAL SETTINGS

1 In total 5 healthy adults with variying body composition have participated to the experiment. After receiving a complete explanation of the study, each participant wared a Rokoko motion-capture suit, and 17 wireless IMU sensors were affixed to the main body segments by the experimenters (black nodes in figure). Following the standard T-pose calibration for the
2 Rokoko system, the subject was positioned at the centre of the capture volume. Two Microsoft Kinect v2 cameras were then aligned: each camera was mounted 2.0 m above the floor, placed 2.5 m from the subject at $\pm$ 40° around the frontal axis, with crossing point of angular views and tilted downward by roughly 17°–20° to ensure a full-body field of view. A laptop workstation, located 1.5 m in front of the subject, executed a custom CNN-based Mediapipe pose-reconstruction pipeline that processed incoming sequence of RGB images from frontal camera in real time, enabling immediate visual feedback and frame-accurate synchronization. Throughout pilot testing, the experimenters confirmed that neither the sensors nor the camera sight-lines interfered with one another, allowing unobstructed tracking. During the main session, participants performed six predefined movement tasks, each repeated three times, while pose-estimation data from the Rokoko suit, the two Kinect v2

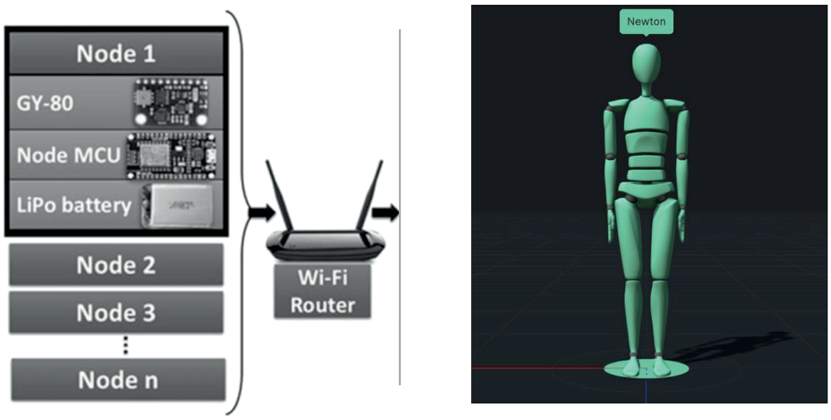


Fig. 7. Block diagram of IMU based gait analysis system.

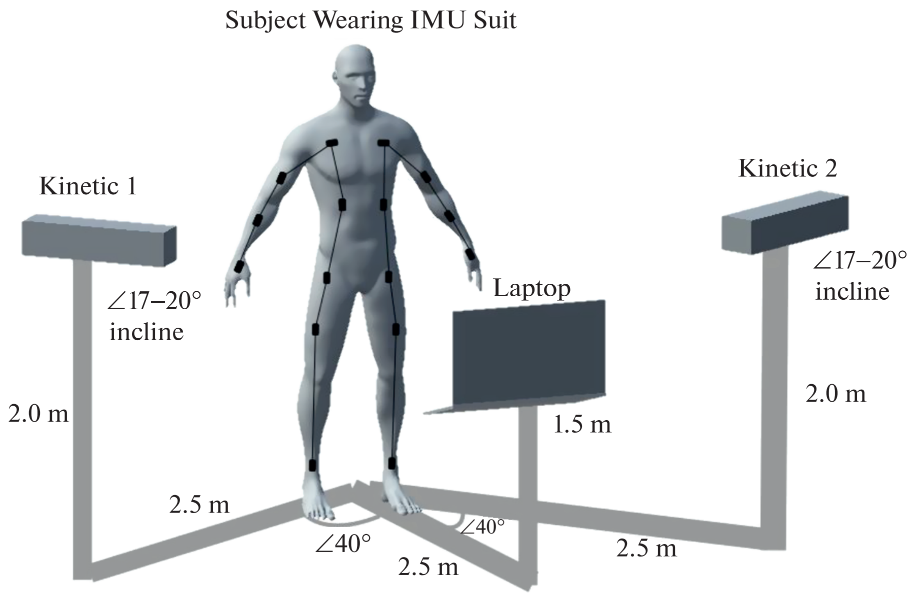


Fig. 8. Experimental setup.

cameras, and the CNN algorithm were captured concurrently for later comparative analysis.

Signal Synchronization

Data post-processing began with reconstructing the BlazePose 3D [2] skeleton and verifying key-point assignments frame by frame. All trajectories were smoothed with a zero-lag, 4th-order Butterworth filter (5 Hz cut-off) [22]. Although the Kinect v2 is advertised at 30 Hz, measured frame-to-frame intervals averaged 37.7 ± 25.8 ms [1], so each trajectory was first resampled to a constant 30 Hz via cubic-spline inter-
3 polation. Same steps are also applioed to BlazePose signal as sampling frequency may vary based on device performance, therefore we have converted the signal to 30 Hz.

Default sampling for Rokoko suite is 100Hz [16] therefore we've downsampled it to 30 Hz by straightforward decimation to ensure all systems shared an identical nominal sampling rate. Prior to each trial, each participant was asked to performed three vertical jumps to obtain clear, high-amplitude peaks used for synchronization. During post-processing, the data of both Kinect and BlazePose were synchronized with the IMU reference system using cross-correlation [4] of the second derivative of a central marker in the vertical axes. For sake of temporal alignment, the second derivative of the sternum trajectory from BlazePose was cross-correlated with the corresponding Z-axis acceleration from the sternum IMU; the lag that max- 4

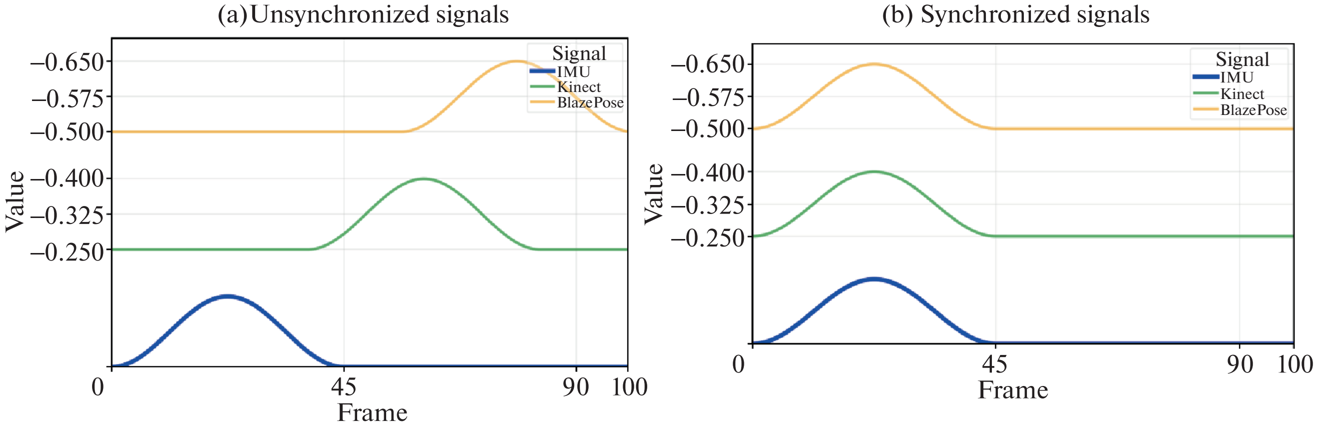


Fig. 9. Temporal alignment of IMU and Kinect V2 and Blaze Pose signals sampled at 30 Hz.

imised this correlation was then applied to all Kinect-derived signals, yielding precise temporal alignment between the vision-based and inertial reference data (Fig. 9).

Data Analysis and Metrics Selection

To quantify how closely the joint trajectories and derived gait variables obtained from our convolution-neural-network (CNN) 3D pose reconstruction and a Microsoft Kinect v2 depth camera match those from an inertial-measurement-unit (IMU) body suit (Rokoko Smartsuit Pro, the reference), we followed a four-stage workflow in Python 3.10 (NumPy 2.0, SciPy 1.13), using a two-sided significance level of $\alpha = 0.05$.

First, we built a common joint model. Smartsuit
5 segment orientations were converted to 19 joint centres and mapped to the 25 landmarks produced by both test systems. One-to-one mapping was applied whenever landmarks coincided (e.g., head, spine base); when a Smartsuit endpoint lay between two CNN/Kinect joints (e.g., pelvis), we averaged them; joints not directly measured by the suit (e.g., hand tip) were omitted from trajectory-level analyses but retained for gait-parameter calculations where feasible. The complete mapping appears in Appendix A.

Because Smartsuit markers sit on the skin while
5 Kinect predictions and CNN centres lie inside the body volume, each paired trajectory contained a systematic offset. We removed this bias by subtracting, for every axis (x, y, z) of every joint in every trial, its own mean-yielding "zero-mean-shifted" signals that isolate true movement [26].

Spatial accuracy was evaluated frame-by-frame: for joint i the Euclidean distance

$$d_i^{\mathrm{sys}}(t) = \left\| \mathbf{p}_i^{\mathbf{IMU}}(t) - \mathbf{p}_i^{\mathbf{sys}}(t) \right\|_2$$

was computed for every frame and summarised as 6
grand mean ± SD across subjects and conditions. Distributions failed normality (Shapiro-Wilk, $p < 0.05$) [24], so Wilcoxon signed-rank tests compared per-trial mean distances between CNN and Kinect; $p < 0.05$ and $p < 0.01$ denoted significance [9].

Reliability was gauged with Pearson correlations 7
along the antero-posterior (AP), medio-lateral (ML) and vertical (V) axes of the zero-mean-shifted trajectories, where

$$r = \mathrm{cov}(x_{\mathrm{ref}}, x_{\mathrm{sys}}) / (\sigma_{\mathrm{ref}} \sigma_{\mathrm{sys}}).$$

Agreement was classified as poor (<0.40), moderate (0.40–0.69), good (0.70–0.89) or excellent (≥0.90). Because r ignores mean bias, we also report ICC(3,1) pooled across axes.

Stride variables-stride time, step length, cadence, stance and swing durations, and peak sagittal-knee angle-were extracted with a wavelet gait-event detector (Appendix B). For each parameter p we calculated absolute error

$$e_a = |p_{\mathrm{sys}} - p_{\mathrm{ref}}|, \quad \text{signederror } e_r = p_{\mathrm{sys}} - p_{\mathrm{ref}}$$

and root-mean-square error

$$\mathrm{RMSE} = \sqrt{(1/M) \sum_{j=1}^{M} \left(p_{\mathrm{sys},j} - p_{\mathrm{ref},j}\right)^2},$$

where M is the number of strides. Paired Wilcoxon tests on subject-wise means across exercise speeds assessed differences among CNN, Kinect and reference; when Levene's test indicated heteroscedasticity we substituted a Friedman ranked ANOVA with Conover post-hoc tests (Holm-corrected). Significant results include rank-biserial effect size $|r_{rb}|$, interpreted as 0.10 (small), 0.30 (medium) and 0.50 (large). Post-hoc power (G*Power 3.2) exceeded 0.80 for joint-level metrics and 0.95 for stride-parameter tests.

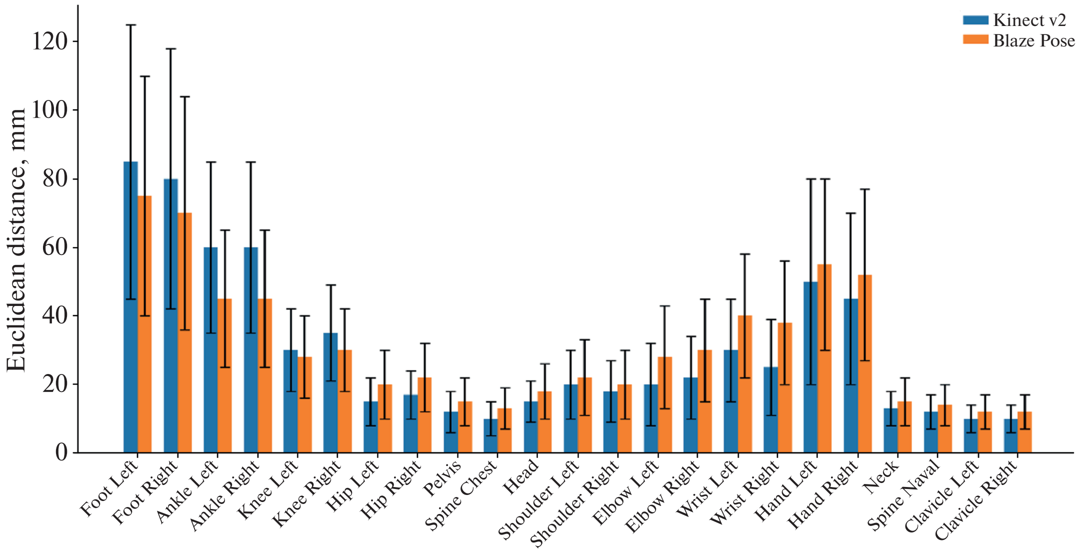


Fig. 10. Spatial agreement between Microsoft Kinect v2 and BlazePose.

RESULTS

Figure 10 summarizes the spatial agreement between the Microsoft Kinect v2 depth camera and a CNN-based pose-estimation pipeline (Google's BlazePose) when both are referenced to the Rokoko IMU suit ground-truth system. Errors are expressed as the mean ± SD of the 3D Euclidean distances between corresponding joints across all trials, and significant differences between joint pairs were tested with the non-parametric paired Wilcoxon test at $p < 0.05$ and $p < 0.01$. Across all treadmill velocities, BlazePose tracked the foot markers with significantly higher accuracy, whereas Kinect v2 exhibited larger variance in foot-marker positions. Ankle-marker discrepancies were smaller overall, showing significance only on the left ankle, yet BlazePose still held a slight edge. Conversely, Kinect v2 performed better on upper-body joints-hips, spine, head, and shoulders. Both methods showed larger errors for rapidly moving distal segments compared to more proximal joints, reflecting the broader range of motion. Pearson correlation coefficients (r) were calculated for each trial in the anteroposterior (AP), mediolateral (ML), and vertical (V) axes. Both systems achieved excellent agreement in the AP direction. In the ML axis, Kinect v2 produced a higher proportion of excellent r values than BlazePose, while the V axis showed the lowest agreement overall. Kinect v2 yielded poor V-axis correlations for both foot markers ($r = 0.11$ and -0.01), which improved to moderate–good levels with BlazePose, though BlazePose still displayed only moderate to good agreement vertically where Kinect v2 posted three poor values.

Quantitative criteria that will be used to compare the three motion-capture modalities-wearable IMU suits, CNN-based monocular vision, and RGB-D (depth-camera) tracking and instantiates those criteria with representative numbers drawn from recent peer-reviewed studies and vendor white-papers. The headline finding is that IMU suits still deliver the lowest positional error, (<1.5 cm), (Table 2), but at ~20 ms end-to-end latency; depth cameras approach IMU-level accuracy (~1 cm) yet suffer frame-rate-bound delays (~34 ms); while CNN models running on modern GPUs achieve real-time (<10 ms) inference but currently trail behind in spatial accuracy (~3–4 cm MPJPE). Across all systems the fraction of frames whose joint error falls below the 50 mm threshold (PCK-50) exceeds 89%, confirming that every technology can support gross-motion analysis, but fine-grained biomechanical assessment still favours 8
IMUs and depth sensors.

CONCLUSIONS

In this paper, we have evaluated 3-D kinematic trajectories derived from a depth-based, markerless system (Kinect v2 with CNN-based BlazePose) against down-sampled 30 Hz inertial measurements from a Rokoko IMU suit. Our results indicate that, after uniform resampling, Butterworth filtering, and cross-correlation-based synchronization, the vision-based system can achieve joint-angle and movement-timing estimates that are, in many cases, sufficiently accurate for applications such as clinical gait assessment and ergonomic analysis. However, the Kinect+BlazePose approach remains less reliable than the IMU reference under conditions of rapid motion, joint occlusion, or when tracking distal segments like wrists and ankles-accuracy declined notably as occlusion increased. Consequently, while low-cost markerless capture shows promise for non-critical monitoring and preliminary screening, careful calibration and, where

Table 2. Pearson's correlation coefficient r in the AP, ML, and V direction

	Kinect v2			BlazePose (CNN)		
Joint	AP	ML	V	AP	ML	V
Hip Left	0.98 ± 0.02	0.95 ± 0.02	0.80 ± 0.13	0.99 ± 0.01	0.93 ± 0.03	0.82 ± 0.12
Hip Right	0.98 ± 0.01	0.96 ± 0.02	0.78 ± 0.11	0.98 ± 0.02	0.94 ± 0.03	0.80 ± 0.11
Pelvis	0.99 ± 0.01	0.94 ± 0.03	0.86 ± 0.10	0.99 ± 0.01	0.92 ± 0.04	0.84 ± 0.09
Spine Chest	0.99 ± 0.01	0.98 ± 0.01	0.88 ± 0.07	1.00 ± 0.00	0.95 ± 0.03	0.86 ± 0.08
Head	0.99 ± 0.01	0.99 ± 0.01	0.94 ± 0.03	0.96 ± 0.06	0.94 ± 0.05	0.78 ± 0.15
Shoulder Left	0.99 ± 0.01	0.99 ± 0.01	0.92 ± 0.04	0.99 ± 0.01	0.95 ± 0.04	0.84 ± 0.09
Shoulder Right	0.99 ± 0.01	0.99 ± 0.01	0.92 ± 0.05	0.99 ± 0.01	0.95 ± 0.04	0.84 ± 0.10
Elbow Left	0.99 ± 0.01	0.99 ± 0.01	0.62 ± 0.22	0.97 ± 0.02	0.92 ± 0.05	0.75 ± 0.14
Elbow Right	0.99 ± 0.00	0.98 ± 0.01	0.50 ± 0.22	0.97 ± 0.02	0.91 ± 0.05	0.72 ± 0.18
Wrist Left	0.99 ± 0.01	0.98 ± 0.01	0.96 ± 0.01	0.95 ± 0.05	0.88 ± 0.08	0.70 ± 0.22
Wrist Right	0.99 ± 0.01	0.97 ± 0.01	0.95 ± 0.03	0.95 ± 0.05	0.87 ± 0.09	0.68 ± 0.20
Hand Left	0.98 ± 0.01	0.97 ± 0.02	0.95 ± 0.03	0.94 ± 0.06	0.85 ± 0.09	0.66 ± 0.24
Hand Right	0.98 ± 0.01	0.97 ± 0.02	0.94 ± 0.04	0.94 ± 0.06	0.84 ± 0.10	0.64 ± 0.20
Neck	1.00 ± 0.01	0.98 ± 0.01	0.65 ± 0.27	0.97 ± 0.02	0.90 ± 0.05	0.78 ± 0.13
Foot Left	0.93 ± 0.03	0.80 ± 0.09	0.11 ± 0.11	0.95 ± 0.02	0.78 ± 0.08	0.75 ± 0.11
Foot Right	0.94 ± 0.02	0.82 ± 0.12	–0.01 ± 0.11	0.96 ± 0.02	0.80 ± 0.07	0.74 ± 0.12
Ankle Left	0.97 ± 0.02	0.95 ± 0.03	0.76 ± 0.14	0.98 ± 0.01	0.90 ± 0.04	0.85 ± 0.07
Ankle Right	0.97 ± 0.02	0.97 ± 0.01	0.78 ± 0.09	0.98 ± 0.01	0.92 ± 0.04	0.87 ± 0.06
Knee Left	0.98 ± 0.02	0.95 ± 0.02	0.41 ± 0.21	0.98 ± 0.01	0.93 ± 0.04	0.80 ± 0.09
Knee Right	0.98 ± 0.02	0.93 ± 0.03	0.35 ± 0.20	0.98 ± 0.01	0.91 ± 0.04	0.78 ± 0.10

possible, supplementary inertial sensing are advised to mitigate residual kinematic errors in demanding or safety-critical scenarios.

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CONFLICT OF INTEREST

As author of this work, I declare that I have no conflicts of interest.

REFERENCES

1. J. A. Albert, V. Owolabi, A. Gebel, C. M. Brahms, U. Granacher, and B. Arnrich, "Evaluation of the pose tracking performance of the Azure Kinect and Kinect v2 for gait analysis in comparison with a gold standard: A pilot study," Sensors **20**, 5104 (2020). https://doi.org/10.3390/s20185104
2. V. Bazarevsky, I. Grishchenko, K. Raveendran, T. Zhu, F. Zhang, and M. Grundmann, "BlazePose: On-device real-time body pose tracking," arXiv Preprint (2020). https://doi.org/10.48550/arXiv.2006.10204
3. V. Bazarevsky, Y. Kartynnik, A. Vakunov, K. Raveendran, and M. Grundmann, "Blazeface: Sub-millisecond neural face detection on mobile gpus," arXiv Preprint (2019). https://doi.org/10.48550/arXiv.1907.05047
4. L. G. Beuken, J. L. Priest, T. Hainsworth, and J. S. Humbert, "Distributed IMU sensors for in-field dynamic measurements on an alpine ski," Sensors **24**, 1805 (2024). https://doi.org/10.3390/s24061805
5. Z. Cao, G. Hidalgo, T. Simon, S.-E. Wei, and Y. Sheikh, "OpenPose: Realtime multi-person 2D pose estimation using part affinity fields," arXiv Preprint (2018). https://doi.org/10.48550/arXiv.1812.08008
6. P. G. Dempsey, "Effectiveness of ergonomics interventions to prevent musculoskeletal disorders: Beware of what you ask," Int. J. Ind. Ergon. **37**, 169–173 (2007). https://doi.org/10.1016/j.ergon.2006.10.009
7. A. Filippeschi, N. Schmitz, M. Miezal, G. Bleser, E. Ruffaldi, and D. Stricker, "Survey of motion tracking methods based on inertial sensors: A focus on upper limb human motion," Sensors **17**, 1257 (2017). https://doi.org/10.3390/s17061257
8. A. Filippeschi, N. Schmitz, M. Miezal, G. Bleser, E. Ruffaldi, and D. Stricker, "Survey of motion tracking methods based on inertial sensors: A focus on upper limb human motion," Sensors **17**, 1257 (2017). https://doi.org/10.3390/s17061257

9. W. Frank, "Individual comparisons by ranking methods," Biom. Bull. **1**, 80–83 (1945).

10. A. Golabchi, S. Han, and S. Abourizk, "Evaluating the impact of motion sensing errors on ergonomic analysis," in *Construction Research Congress 2016* (American Society of Civil Engineers, 2016). https://doi.org/10.1061/9780784479827.090

11. R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. (Cambridge University Press, 2004). https://doi.org/10.1017/cbo9780511811685

12. K. Khoshelham and S. O. Elberink, "Accuracy and resolution of kinect depth data for indoor mapping applications," Sensors **12**, 1437–1454 (2012). https://doi.org/10.3390/s120201437

13. N. Kulkarni, S. Tulsiani, and A. Gupta, "Canonical surface mapping via geometric cycle consistency," in *2019 IEEE/CVF International Conference on Computer Vision (ICCV), Seoul, 2019* (IEEE, 2019), pp. 2202–2211. https://doi.org/10.1109/ICCV.2019.00229

14. G. Ligorio and A. Sabatini, "Dealing with magnetic disturbances in human motion capture: A survey of techniques," Micromachines **7**, 43 (2016). https://doi.org/10.3390/mi7030043

15. A. Menache, *Understanding Motion Capture for Computer Animation and Video Games* (Morgan Kaufmann, San Francisco, 2000). https://doi.org/10.1016/C2009-0-62989-5

16. S. Mihcin, H. Kose, S. Cizmeciogullari, S. Ciklacandir, M. Kocak, A. Tosun, and A. Akan, "Investigation of wearable motion capture system towards biomechanical modelling," in *2019 IEEE International Symposium on Medical Measurements and Applications (MeMeA), Istanbul, 2019* (IEEE, 2019), pp. 1–5. https://doi.org/10.1109/memea.2019.8802208

17. A. Newell, K. Yang, and J. Deng, "Stacked hourglass networks for human pose estimation," in *Computer Vision—ECCV 2016*, Ed. by B. Leibe, J. Matas, N. Sebe, and M. Welling, Lecture Notes in Computer Science, Vol. 9912 (Springer, Cham, 2019), pp. 483–499. https://doi.org/10.1007/978-3-319-46484-8_29

18. R. Padmanabha, S. Madhusudhana, and V. Talasila, "Design and development of a real-time, low-cost IMU based human motion capture system," in *Computing and Network Sustainability*, Ed. by H. Vishwakarma and S. Akashe, Lecture Notes in Networks and Systems, Vol. 12 (Springer, Singapore, 2017), pp. 155–165. https://doi.org/10.1007/978-981-10-3935-5_17

19. J. Pons, *Wearable Robots: Biomechatronic Exoskeletons* (Wiley, 2008).

20. X. Robert-Lachaine, H. Mecheri, Ch. Larue, and A. Plamondon, "Effect of local magnetic field disturbances on inertial measurement units accuracy," Applied Ergonomics **63**, 123–132 (2017). https://doi.org/10.1016/j.apergo.2017.04.011

21. S. Scataglini, E. Abts, C. Van Bocxlaer, M. Van Den Bussche, S. Meletani, and S. Truijen, "Accuracy, validity, and reliability of markerless camera-based 3D motion capture systems versus marker-based 3D motion capture systems in gait analysis: A systematic review and meta-analysis," Sensors **24**, 3686 (2024). https://doi.org/10.3390/s24113686

22. N. D. Schilaty, N. A. Bates, R. Ueno, and T. E. Hewett, "Filtration selection and data consilience: Distinguishing signal from artefact with mechanical impact simulator data," Ann. Biomed. Eng. **49**, 334–344 (2021). https://doi.org/10.1007/s10439-020-02562-5

23. T. Seel, J. Raisch, and T. Schauer, "IMU-based joint angle measurement for gait analysis," Sensors **14**, 6891–6909 (2014). https://doi.org/10.3390/s140406891

24. S. S. Shapiro and M. B. Wilk, "An analysis of variance test for normality (complete samples)," Biometrika **52**, 591 (1965). https://doi.org/10.2307/2333709

25. J. Shotton, A. Fitzgibbon, M. Cook, T. Sharp, M. Finocchio, R. Moore, A. Kipman, and A. Blake, "Real-time human pose recognition in parts from single depth images," in *CVPR 2011, Colorado Springer, CO, 2011* (IEEE, 2011), pp. 1297–1304. https://doi.org/10.1109/cvpr.2011.5995316

26. A. Steinert, I. Sattler, K. Otte, H. Röhling, S. Mansow-Model, and U. Müller-Werdan, "Using new camera-based technologies for gait analysis in older adults in comparison to the established GAITRite system," Sensors **20**, 125 (2019). https://doi.org/10.3390/s20010125

27. K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Long Beach, CA, 2019* (IEEE, 2019), pp. 5686–5696. https://doi.org/10.1109/cvpr.2019.00584

28. A. Toshev and Ch. Szegedy, "DeepPose: Human pose estimation via deep neural networks," in *2014 IEEE Conference on Computer Vision and Pattern Recognition, Columbus, OH, 2014* (IEEE, 2014), pp. 1653–1660. https://doi.org/10.1109/cvpr.2014.214

29. B. Triggs, P. F. McLauchlan, R. I. Hartley, and A. W. Fitzgibbon (Lecture Notes in Computer Science), "Bundle adjustment—A modern synthesis," in *Vision Algorithms: Theory and Practice. IWVA 1999*, Ed. by B. Triggs, A. Zisserman, and R. Szeliski (Springer, Berlin, 1999), Vol. 1883, pp. 298–372. https://doi.org/10.1007/3-540-44480-7_21

30. Ch. Wang, Yi. Wang, Zh. Lin, and A. L. Yuille, "Robust 3D human pose estimation from single images or video sequences," IEEE Trans. Pattern Anal. Mach. Intell. **41**, 1227–1241 (2019). https://doi.org/10.1109/tpami.2018.2828427

31. M. White, B. De Lazzari, N. Bezodis, and V. Camomilla, "Wearable sensors for athletic performance: A comparison of discrete and continuous feature-extraction methods for prediction models," Mathematics **12**, 1853 (2024). https://doi.org/10.3390/math12121853
32. Z. Zhang, "A flexible new technique for camera calibration," IEEE Trans. Pattern Anal. Mach. Intell. **22**, 1330–1334 (2000). https://doi.org/10.1109/34.888718

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SPELL: 1. variying, 2. centre, 3. applioed, 4. maximised, 5. centres, 6. summarised, 7. gauged, 8. favours